

(c) Glycolysis

(d) Hydrolysis

I.A.S. (Pre) 1998

Ans. (a)

Aerobic respiration is the release of energy from glucose or another organic substrate in the presence of oxygen, resultant glucose is converted into CO_2 and H_2O . By this process, large amount of energy is released. Aerobic respiration happens all the time in the cells of animals and plants.

10. Which of the following statements is/are true in reference to break-down of glucose?

(i) In cytoplasm, glucose breaks down into pyruvate which has four carbon molecules.

(ii) Due to lack of oxygen, pyruvate gets converted to lactic acid.

(iii) In absence of oxygen, pyruvate gets converted to carbon dioxide and ethanol.

(a) (i), (ii) and (iii)

(b) (i) and (iii)

(c) (ii) and (iii)

(d) Only (iii)

Chhattisgarh P.C.S. (Pre) 2021

Ans. (c)

In the process of cellular respiration, glucose breaks down into pyruvate through glycolysis in cytoplasm. Pyruvate (CH_3COCO^-) has three carbon molecules. In presence of oxygen (in aerobic respiration) pyruvate produces carbon dioxide and water, along with the generation of energy. Due to lack of oxygen, pyruvate gets converted to lactic acid in the muscles. In absence of oxygen (in anaerobic respiration), pyruvate gets converted to carbon dioxide and ethyl alcohol (ethanol).

11. Which part of human body maintains the body temperature?

(a) Heart

(b) Lungs

(c) Liver

(d) Kidney

M.P.P.C.S. (Pre) 1991

Ans. (b)

The lungs are main respiratory organ, which maintains the body temperature through breathing, inhaling oxygen-rich air and exhaling air filled with carbon dioxide which is a waste gas. In this process, energy is produced in tissues by oxidation of foods and the water of the body is vaporised, which maintains the body temperature.

V. Circulatory System

Notes

- The circulatory system is a network consisting of **blood**, **blood vessels** and the **heart**. This network supplies tissues in the body with oxygen and other nutrients, transport hormones and removes unnecessary waste products.
- There are two main types of circulatory systems : **open circulatory systems** and **closed circulatory systems**.
- Open circulatory systems** are systems where internal organs and body tissues are surrounded by circulatory fluid, e.g. in crustaceans, insects, molluscs and other invertebrates.
- Closed circulatory systems** have the blood closed at all times within vessels of different size and wall thickness. In this type of system, blood is pumped by a heart through vessels and does not normally fill body cavities. Large and more active animals, including all vertebrates, have a closed circulatory system.

Human Heart :

- The human heart is a muscular organ, which pumps blood throughout the body via the blood vessels of the circulatory system, supplying oxygen and nutrients to the tissues and removing carbon dioxide and other wastes.
- In human, the heart is roughly the size of a large fist and weighs between about 250-300 grams in adults.
- The human heart is located between the lungs, in the middle compartment (mediastinum) of the chest.
- The **Pericardium** is a double-walled sac enclosing the heart, consisting of an outer **fibrous membrane** and an inner double layer of **serous membrane**. The intervening space between these layers is filled with pericardial fluid which protects the heart from outer shocks and acts as a lubricant to allow normal heart movement within the chest.
- The wall of heart is made up of three layers : epicardium, myocardium, and endocardium.
- The human heart has four chambers- two upper chambers (the atria or auricles - receiving chambers) and two lower ones (the ventricles-discharging chambers).
- The right atrium and right ventricle together make up the 'right heart' and the left atrium and left ventricle make up the 'left heart'. A wall of muscle called the **septum** separates the two sides of the heart.

- In a healthy heart, blood flows one way through the heart due to heart valves which prevent backflow.
- The heart pumps blood with a rhythm determined by a group of pacemaking cells in the **sinoatrial node**. These generate a current that causes contraction of the heart, travelling through the **atrioventricular** node and along the conduction system of the heart. The heart receives blood low in oxygen from the system circulation, which enters the right atrium from the superior and inferior **vena cava** and passes to the right ventricle. From there it is pumped into the **pulmonary circulation**, through the lungs where it receives oxygen and gives off carbon dioxide. Oxygenated blood then returns to the left atrium, passes through the left ventricle and is pumped out through the **aorta** to the systemic circulation— where the oxygen is used and metabolized to carbon dioxide.
- The heart beats at a resting rate close to 72-75 beats per minute.
- **Trachycardia** is considered a heart rate of greater than 100 per minute. This may be due to stress, medications or it may be a sign of serious heart condition.
- The **cardiac** cycle is the sequence of events that occurs when the heart beats. As the heart beats, it circulates blood through **pulmonary** and **systemic circuits** of the body. There are two phases of the cardiac cycle. In the **diastole phase**, the heart **ventricles** are relaxed and the heart refills with blood. In the **systole phase**, the ventricles contract and pump blood out of the heart and to **arteries**. One cardiac cycle is completed when the heart chamber fills with blood and blood is then pumped out of the heart.
- **Double circulation** : During a single cardiac cycle blood goes twice in the heart. It is necessary in human being to separate oxygenated and deoxygenated blood because this makes their circulatory system more efficient and helps in maintaining body temperature.
- **William Harvey** discovered the blood circulation system firstly.
- **Dr. Christian Barnard** was a South African cardiac surgeon who performed the world's first highly publicized heart transplant and the first one in which the patient regained consciousness (1967).
- **Natural Pacemaker** : Electrical impulses from the heart muscles cause to beat. This electrical signal begins in the sinoatrial node (S-A node) located at the top of the heart's upper right chamber (the right atrium). The S-A node is a **natural pacemaker**.
- **Artificial Pacemaker** : It is a medical device which uses electrical impulses, delivered by electrodes contacting the heart muscles, to regulate the beating of the heart. The most common reasons to need a pacemaker are **bradycardia** and **heart block**. Bradycardia is a heartbeat that is slower than normal. Heart block is a disorder that occurs if an electrical signal is slowed or disrupted as it moves through the heart. A pacemaker can help to regulate heartbeat.
- **Jarvik-7 (Artificial heart)** : A mechanical pump used to replace the function of a damaged heart, either temporarily or as a permanent prosthesis. It was invented by **Robert Jarvik**. **Barny Clark** was the first recipient of an artificial heart in 1982.

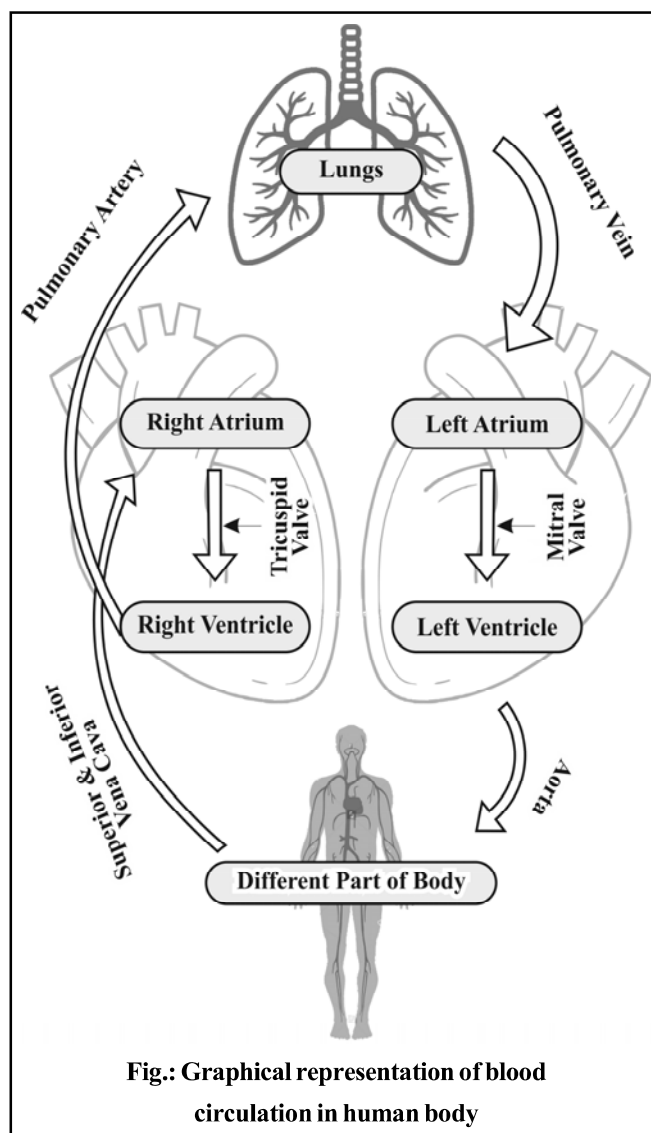


Fig.: Graphical representation of blood circulation in human body

Blood Pressure :

- Blood pressure is the pressure of blood on the walls of arteries as the heart pumps it around the body. It is a vital part of how heart and blood circulation works.
- The normal blood pressure of human is **120/80 mm Hg**. The top number refers to the amount of pressure in arteries during the contraction of the heart muscle. This is called **systolic pressure**. The bottom number refers to blood pressure when heart muscle relaxes. This is called **diastolic pressure**.
- **High blood pressure** is when blood pressure is persistently higher than normal.
- Blood pressure that's high over a long time is one of the main risk factors for heart disease. As humans get older, the chances of having persistently high blood pressure increases.
- It is very important to get blood pressure checked regularly, and if it is persistently high it needs to be controlled.
- Uncontrolled high blood pressure can lead to a heart attack or stroke. It may attack kidneys.
- The medical name for high blood pressure over a long period of time is **hypertension**.
- The causes of high blood pressure are as follows-
 - family history
 - eating patterns including salty foods
 - alcohol and tobacco intake
 - weight gain
 - low physical activity
 - mental tension
- **Low blood pressure** is when blood pressure is much below than the normal range. In this condition flow of blood to the organs of the body is inadequate. The medical term for low blood pressure is **hypotension**.
- Low blood pressure alone, without symptoms or signs, usually is not unhealthy.
- The symptoms of low blood pressure include light-headedness, dizziness, blurred vision and fainting. These symptoms are most prominent when individuals go from the lying or sitting position to the standing position (Orthostatic hypotension).
- Low blood pressure that causes an inadequate flow of blood to the body's organs can cause strokes, heart attacks and kidney failure. The most severe form of low blood pressure is **shock**.
- Common causes of low blood pressure include a reduced volume of blood, dehydration, heart disease and medications.

- The blood pressure is measured by an apparatus called **sphygmomanometer**.

Blood Vessels :

- The blood vessels are part of the circulatory system and function to transport blood throughout the body.
- The most important types, **arteries** and **veins**, carry blood away from or towards the heart respectively.
- Around this, there is a layer of vascular smooth muscle, which is highly developed in arteries.
- **Semilunar valves** are found in **veins** which only permit the flow of blood towards the heart and not in the opposite direction. No such valve is found in arteries.
- Arteries from the heart branch into small passages called **arterioles** and then into the **capillaries**. Capillaries merge into **venules** (to bring blood into the venous system), which merge into veins leading back to the heart.
- In humans, the **hepatic portal vein** combines from capillaries around the gastrointestinal tract where the blood absorbs the various products of digestion, and rather than leading directly back to heart, the hepatic portal vein branches into a second capillary system in the liver.

Blood :

- It is a fluid, a specialized form of **connective tissue**.
- Its pH is about 7.4 (alkaline solution).
- The amount of blood in the human body is generally equivalent to about 7% of body weight. Its average density around 1060 kg/m³. The average adult has a blood volume of roughly 5-6 litres.
- Blood is essential to life. It circulates through the body and delivers essential substances like oxygen and nutrients to the body's cells. It also transports metabolic waste products away from those cells. There is no substitute for blood. It cannot be made or manufactured. Generous blood donors are the only source of blood for patients in need of a blood transfusion.

Components of the blood :

- The main components of the blood are –
 - (A) Plasma
 - (B) Blood cells or corpuscles

(A) Plasma :

- Plasma is the relatively clear, yellow tinted water (about 91-92%), sugar, fat, protein and salt solution which carries the blood corpuscles.

- Normally about 55% of our blood volume is made of plasma.
- Plasma also contains blood clotting factors, vitamins, enzymes, hormones, antibodies and other proteins.

(B) Blood corpuscles :

- There are three types of blood corpuscles in human blood :

(a) Red Blood Corpuscles (Erythrocytes)

(b) White Blood Corpuscles (Leucocytes)

(c) Blood Platelets (Thrombocytes)

(a) Red Blood Corpuscles (Erythrocytes) :

- In humans and other mammals, mature red blood cells are flexible and oval biconcave disks. They lack a nucleus (exception – the RBCs of **Camel** and **Lama**) and most organelles, in order to accommodate maximum space for haemoglobin.
- Approximately 2.4 million new erythrocytes are produced per second in human adults.
- The red blood cells develop in the **bone marrow** of the human bones (in the liver in embryo) and circulate for about 100-120 days in the body before their components are recycled by macrophages. Each circulation takes about 60 seconds.
- In human body, nearly half of the blood volume (about 40% to 45%) is red blood cells.
- The number of RBCs in male human is about 55 lakh per microlitre (cubic mm) of blood; in female about 50 lakhs per microlitre and in person of the hilly area 70-75 lakhs per microlitre.
- Spleen is the **blood bank of the body** and old red blood cells are recycled in the spleen.
- Haemoglobin is the iron-containing **oxygen-transport metalloprotein** in the red blood cells of almost all vertebrates (the exception being the fish family **Channichthyidae**) as well as the tissues of some invertebrates.
- Haemoglobin in blood carries oxygen from the **lungs** or **gills** to the rest of the body (i.e. tissues). There it releases the oxygen to permit aerobic respiration to provide energy to power the functions of the organism in the process called **metabolism**.
- A healthy person has about 12 to 18 grams of haemoglobin in every 100 ml. of blood.
- **Anaemia** means that either the level of RBCs or the level of haemoglobin is lower than normal in the blood. When a person has anaemia, his heart has to work harder to pump

the quantity of blood needed to get enough oxygen around their body.

- **Polycythemia**, or erythrocytosis, is a disease state in which the volume percentage of red blood cells in the blood is elevated, which causes blood to get too thick. It can be due to an increase in the number of RBCs or to a decrease in the volume of plasma. That can make you more likely to have clots, a stroke or heart attack.

- The haemoglobin is also known as **respiratory pigment**.

(b) White Blood Corpuscles (Leucocytes) :

- White blood corpuscles (WBCs) are the cells of the immune system that are involved in protecting the body against both infectious diseases and foreign invaders.
- All WBCs are produced and derived from multipotent cells in the bone marrow known as hematopoietic stem cells.
- All WBCs have nuclei, which distinguishes them from the other blood cells (RBCs and Platelets).
- White blood cells make up merely about 0.7-1% of the total blood volume in a healthy adult, but body's immunity depends on it.
- Normally white blood cell's number is near about 4000-11000 per microlitre of blood.
- The life of white blood cells is generally a few hours to a few days.
- The WBCs have been classified into two classes –
 - (i) Granulocytes
 - (ii) Agranulocytes
- (i) **Granulocytes** – Their Cytoplasm contains granules. Granulocytes have been divided in 3 parts as follows-
 1. **Eosinophils** : Eosinophils are specialized cells of the immune system. This pro-inflammatory white blood cells have a nucleus with two lobes and cytoplasm filled with about 200 large granules containing enzymes and proteins. These are implicated in numerous inflammatory processes, especially in allergic disorders.
 - Eosinophils compose about 2-4% of the WBCs.
 - **Eosinophilia** is a condition in which the eosinophil count in the blood exceeds. The symptoms of eosinophilia are wheezing and breathlessness.
 2. **Basophils** : Basophils contain anticoagulant heparin (a carbohydrate) which prevents blood from clotting in blood vessels. They also contain the vasodilator **histamine**, which promotes blood flow to tissues.
 - Basophils are the least common of the granulocytes, representing about 0.5 to 1% of circulating WBCs.
 - Basophils contain large cytoplasmic granules. The nucleus usually has two lobes.

3. **Neutrophils** : Neutrophils are also known as the most abundant type of granulocytes and the most abundant (60% to 65%) type of WBCs. They form an essential part of the innate immune system. Neutrophils help heal damaged and resolve infections.

- Neutrophils levels in blood increase naturally in response to infections, injuries and other types of infections.
- Neutrophils contain a nucleus divided into 2-5 lobes.

(ii) **Agranulocytes** – Agranular cells are noted by the absence of granules in their cytoplasm. They are of two types :

1. Lymphocytes, 2. Monocytes.

1. **Lymphocytes** : Lymphocytes are one of the body's main types of immune cell. The two main types of lymphocytes are B lymphocytes and T lymphocytes. B lymphocytes make antibodies and T lymphocytes help kill tumour cells and help control immune responses.

- T lymphocyte cells and B lymphocyte cells derive their names from the organs in which they develop. T cells develop in the thymus, and B cells, in mammals, develop in the bone marrow in adults (or the liver in fetuses).

- In mature individuals, development of new T cells in the thymus slows down and T-cell numbers are maintained through division of mature T cells in spleen and lymph nodes. New B cells, on the other hand, are continually produced from the bone marrow.

2. **Monocytes** : A type of immune cell that is made in the bone marrow and travels through the blood to tissues in the body where it becomes a **macrophage**. Macrophages surround and kill microorganisms, ingest foreign material, remove dead cells and boost immune system. Monocytes are the largest in all WBCs.

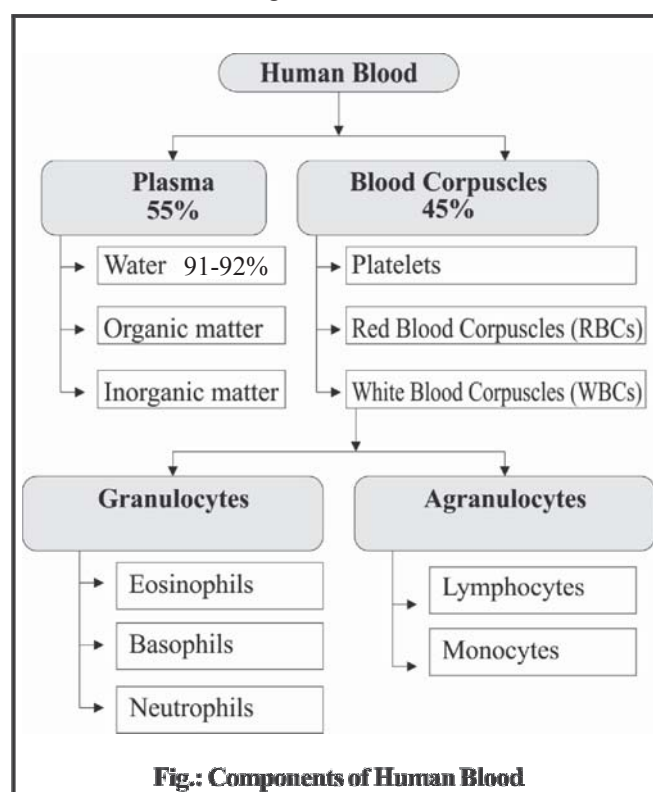
- **Leukaemia** is a cancer of the **blood** or **bone marrow**. It leads to the over-production of abnormal white blood cells – the part of the immune system which defends the body against infection.

(c) **Blood Platelets (Thrombocytes)** :

- Platelets are tiny blood cells formed in the bone marrow.
- Platelets have no cell nucleus.
- They generally live for almost 8 to 10 days.
- A normal platelet count is 1,50,000 to 4,50,000 per microlitre.
- The risk for spontaneous bleeding develops if a platelets count falls below 10,000 to 20,000. But when the platelet count is less than 50,000, bleeding is likely to be more serious if an individual is cut or bruised.
- Platelets help body in forming clots to stop bleeding.

Production of Blood Cells :

- In the human adult, the bone marrow alone produces all of the RBCs, 60-70 percent of the WBCs (i.e., the granulocytes), and all of the platelets.
- The bone marrow and the lymphatic tissues, particularly the thymus, the spleen, and the lymph nodes, produce the lymphocytes (20-25 percent of the WBCs).
- Monocytes (4-8 percent of the WBCs) are produced by the reticuloendothelial tissues of the spleen, liver, lymph nodes, and other organs.



Blood Group :

- Blood Group was discovered by **Landsteiner** in 1900.
- Blood Group is a classification of blood, based on the presence and absence of **antibodies** and inherited **antigenic** substances. The antigen is present on the surface of RBC while antibody is present in the plasma.
- The antigen is a glycoprotein. It is of two types, antigen-A and antigen-B.
- In the absence of antigen, a special type of protein is found in plasma known as antibodies. It is of two types, antibody-a and antibody-b.
- On the above basis, the human blood group has been classified into four groups.
- Chart of Blood Group is given below :

Blood Group	Antigen	Antibody	Remarks
A	A	b	–
B	B	a	–
AB	A, B	–	Universal recipient
O	–	a, b	Universal Donor

- There is an **agglutination** reaction between similar antigen and antibody (for example, antigen-A agglutinates the antibody-a and antigen-B agglutinates the antibody-b). Thus transfusion can be considered safe as long as the serum of the recipient does not contain antibodies for the blood cell antigens of the donor.
- Due to the absence of antigen in blood group O, it is **universal donor** and due to the absence of antibody in blood group AB, it is **universal recipient**.

Blood Transfusion Chart

Donor Group	Recipient Group			
	A	B	AB	O
A	√	×	√	×
B	×	√	√	×
AB	×	×	√	×
O	√	√	√	√

- According to **Bernstein** (1924) blood groups A, B, O are the heritable trait of human which passes to progeny in accordance with Mendel's law of Inheritance.

Blood Group of Parent	Blood Group of Progeny	
	Possible	Impossible
O×O	O	A, B, AB
O×A	O, A	B, AB
A×A	O, A	B, AB
O×B	O, B	A, AB
B×B	O, B	A, AB
A×B	O, A, B, AB	–
O×AB	A, B	O, AB
A×AB	A, B, AB	O
B×AB	A, B, AB	O
AB×AB	A, B, AB	O

Rh-Factor :

- In 1937 **Landsteiner** and **Wiener**, discovered another antigen in blood of monkey **Rhesus**, hence the name **Rh-factor** was given to this antigen.
- Rh-factor is an inherited protein found on the surface of RBCs. If blood has that protein, the blood is Rh-positive and if blood lacks that protein, the blood is Rh-negative.
- Rh-positive is the most common blood type.
- During blood transfusion, the same Rh-factor must be considered to avoid the agglutination.
- With consideration of Rh-factor, the **universal donor** is **O-negative** blood group while the **universal recipient** is **AB-positive** blood group.

Erythroblastosis fetalis :

- Erythroblastosis fetalis, also known as **hemolytic anaemia**, is a serious medical condition that most commonly results from maternal-fetal blood type (Rh-factor) incompatibility.
- If a woman who is Rh-negative and a father who is Rh-positive conceive a child, the fetus may be Rh-positive. This can lead to Rh incompatibility and pregnancy complications..
- If mother's blood and fetus's blood mix, mother's body will start to make antibodies that can damage fetus's blood cells (RBCs). This could cause the baby to develop severe anaemia and other problems.
- Treatment options for newborns with the condition include -
 - blood transfusion
 - intravenous (IV) fluids
 - managing breathing problems
 - IV immunoglobulin

Question Bank

- How many times, a healthy person's average heart beats in a minute–

- (a) 86 times (b) 98 times
(c) 72 times (d) 64 times

Uttarakhand P.C.S. (Pre) 2007
M.P.P.C.S. (Pre) 2000

Ans. (c)

Heart rate is the speed of the heartbeats, measured by the number of contractions of the heart typically beats per minute. Our heart beats about 72-75 times a minute under normal conditions. During sleep a slow heartbeat rates around 40-50 BPM is common. The cardiac cycle refers to a complete heart beat from its generation to the beginning of the next beat. This frequency of the cardiac cycle is described by the heart rate or expressed as beats per minute (BPM).

2. When does heart rest ?

- (a) Never (b) While sleeping
(c) Between two beats (d) While doing yoga

U.P.P.C.S. (Pre) 2008
U.P. Lower Sub. (Pre) 2004

Ans. (c)

Each 'heartbeat' consists of a contraction and relaxing of the heart muscles. When the heart contracts, it pumps blood through the blood vessels. When it relaxes, it draws blood in. It takes rest between relaxation and contraction.

3. Which of the following is not a symptom of a heart attack?

- (a) Chest pain
(b) Nausea and sweating
(c) Numbness and aching of arms
(d) Pain in legs

M.P.P.C.S. (Pre) 2000

Ans. (d)

Heart failure and heart disease do not show the same signs for everyone, especially to women. A heart attack is medically known as a myocardial infarction. The symptoms of heart attack include- pain or discomfort in one or both arms, back, neck, jaws or stomach; shortness of breath with or without chest discomfort and other signs such as breaking out in a cold sweat, nausea or lightheadedness. As with men and women's most common heart attack symptom is chest pain or discomfort.

4. Which of these are typical signs of heart attack ?

- (1) Nausea (2) Severe perspiration
(3) Headache (4) Severe chest pain

Select the correct answer from the given code :

- (a) 1, 2, 3 (b) 2, 3, 4
(c) 1, 3, 4 (d) 1, 2, 4

Uttarakhand P.C.S. (Mains) 2002

Ans. (d)

See the explanation of above question.

5. How many chambers are there in heart?

- (a) 2 (b) 4
(c) 6 (d) 8

M.P.P.C.S. (Pre) 2010

Ans. (b)

In humans, other mammals and birds, the heart is divided into four chambers: upper left and right atria; and lower left and right ventricles. Commonly the right atrium and ventricle are referred together as the right heart and their left counterparts as the left heart. In a healthy heart, blood flows one way through the heart due to heart valves, which prevent backflow.

6. Human heart is made up of how many chambers?

- (a) Only four (b) Only two
(c) Only one (d) Only three

U.P. P.C.S. (Pre) 2023

Ans. (a)

A human heart has two upper and two lower chambers (total four chambers). The upper chambers, the right and left atria, receive incoming blood. The lower chambers, the more muscular right and left ventricles, pump blood out of the heart. The heart valves, which keep blood flowing in the right direction, are gates at the chamber openings.

7. Heart beats are caused by an electrical current that originates in the :

- (a) Brain (b) Blood
(c) Heart (d) Spinal cord

Uttarakhand P.C.S. (Mains) 2002

Ans. (c)

An electrical impulse generator, called the "sinus node", sends signals from the right atrium to trigger the heart beat. Like a natural pacemaker, the electrical current follows a web of pathways through the heart, causing the chambers to squeeze and relax in a steady, rhythmic sequence that draws blood into the heart and pulse it out.

8. The pulse in the human wrist beats –

- (a) Faster than the heart
(b) Slower than the heart
(c) At the same rate as the heart
(d) Independently of the heart

U.P.P.C.S. (Mains) 2010

U.P.P.C.S. (Mains) 2008

Ans. (c)

Pulse rate is defined as the rate at which our heart beats. It is the number of times our heart beats per minute (bpm). By checking our pulse using a pulse rate chart, we can find out how well our heart is working as our general health and fitness levels.

9. The first heart transplant was performed by -

- (a) Dr. William Harvey
(b) Sir F.G. Hopkins
(c) Dr. Louis Pasteur
(d) Dr. Christian Bernard

U.P. Lower Sub. (Pre) 2004

M.P.P.C.S. (Pre) 1994

Ans. (d)

The first heart transplant was performed by Dr. Christian Bernard who was a famous surgeon of South Africa. Barnard performed the first transplant on Louis Washkansky on December 3, 1967 at the Groote Schuur Hospital in Cape Town, South Africa.

10. Who among the following had performed the world's first heart transplant?

- (a) Dr. Venugopal (b) William Harvey
(c) Christian Bernard (d) William Betrick

U.P. P.C.S. (Pre) 2021

Ans. (c)

See the explanation of above question.

11. The first human heart transplant was performed in –

- (a) America (b) England
(c) South Africa (d) France

56th to 59th B.P.S.C. (Pre) 2015

Ans. (c)

See the explanation of above question.

12. What is Jarvik-7?

- (a) Electronic leg (b) Pacemaker
(c) Artificial heart (d) Artificial eye

U.P.P.C.S. (Pre) 2007

Ans. (c)

Robert Jarvik is widely known as the inventor of the first successful permanent artificial heart, the Jarvik-7. In 1982, the first implantation of the Jarvik-7 in patient Barney Clark caught the attention of media around the world.

13. What is the function of a pacemaker ?

- (a) It decreases the heart beat
(b) It regulates heart beat
(c) It increases heart beat
(d) It accelerates blood supply to the heart

U.P. P.C.S. (Pre) 1999

Ans. (b)

A pacemaker is a small device that is placed in the chest or abdomen to help control abnormal heart rhythms. This device uses electrical pulses to promote the heart to beat at a normal rate.

14. The 'Pace Maker' is also known as –

- (a) S.A. Nodes (b) A.V. Nodes
(c) Bundle of His (d) Chordate tensional

U.P.P.C.S. (Mains) 2007

Ans. (a)

The sino-atrial node often abbreviated SA-node is the pacemaker of the heart and is responsible for the initiation of the heart beat. The sino-atrial node is located in the right atrium (right upper chamber) of the heart.

15. The only vein that carries oxygenated blood is :

- (a) Pulmonary vein (b) Cystic vein
(c) Cardiac vein (d) Hepatic Portal vein

U.P.P.C.S. (Pre) 2024

Ans. (a)

The pulmonary veins are the only veins in the body that carry oxygenated blood. They carry oxygenated blood from the lungs to the heart's left atrium. All other veins carry deoxygenated or oxygen-poor blood. Similarly, pulmonary arteries are the only arteries in the body that carry oxygen-poor blood. All the rest of arteries carry oxygenated blood. Deoxygenated blood from the body travels to the lungs, where it picks up oxygen and becomes oxygenated. The oxygenated blood is then carried from the lungs to the heart's left atrium via the pulmonary veins. The blood enters the left atrium, a chamber of the heart. From the left atrium, the blood flows into the left ventricle, the heart's main pumping chamber. The left ventricle pumps the oxygenated blood into the aorta, the body's largest artery. The aorta branches into smaller arteries, which carry the oxygenated blood to all parts of the body. Within the tissues, the arteries branch into even smaller vessels called capillaries, where oxygen is exchanged for carbon dioxide. The deoxygenated blood, now carrying carbon dioxide, returns to the heart through veins. The deoxygenated blood enters the heart's right atrium, completing the circulation cycle. The deoxygenated blood is then pumped from the right ventricle to the lungs via the pulmonary artery, starting the cycle again.

16. The total volume of blood in a normal adult human being is :

- (a) 5-6 litres (b) 3-4 litres
(c) 8-10 litres (d) 10-12 litres

56th to 59th B.P.S.C. (Pre) 2015

Ans. (a)

Blood is a special connective tissue consisting of a fluid matrix, plasma, and formed elements. The total volume of blood in a 70 kg normal adult human being is 5-6 litres.

17. The total amount of blood present in a normal human (70 kg) adult is about :

- (a) 6000 ml (b) 5000 ml
(c) 4000ml (d) 3000 ml

U.P. R.O./A.R.O. (Pre) 2017

Ans. (*)

See the explanation of above question.

18. Blood is –

- (a) Connective tissue
- (b) Epithelial tissue
- (c) Both of the above
- (d) None of the above

Jharkhand P.C.S. (Pre) 2010

Ans. (a)

Blood is a connective tissue that has a fluid matrix, called plasma, and no fibres. It flows inside the blood vessels and is viscous thick fluid. The function of blood is to transport oxygen and carbon dioxide. Actually blood carries away oxygen from lungs towards the body tissues and carries carbon dioxide from there, to take towards lungs.

19. Which among the following help in circulation of blood?

- (a) Lymphocytes
- (b) Monocytes
- (c) Erythrocytes
- (d) Blood platelets
- (e) None of the above/More than one of the above

66th B.P.S.C. (Pre) (Re. Exam) 2020

Ans. (c)

Erythrocytes (RBCs : Red blood cells) help in circulation of blood. RBCs contain haemoglobin, a protein that gives blood its red colour and enables it to carry oxygen from the lungs and deliver it to all body tissues. Oxygen is used by cells to produce energy that the body needs, leaving carbon dioxide as a waste product. RBCs also carry carbon dioxide away from the tissues and back to the lungs.

20. Which of the following is the fluid part of the blood ?

- (a) WBC
- (b) Plasma
- (c) Blood platelets
- (d) RBC

70th B.P.S.C. (Pre) 2024

Ans. (b)

The fluid part of blood is called plasma, a straw-coloured liquid that suspends blood cells and various dissolved substances. Blood is made up of about 55% plasma and about 45% blood cells (red blood cells, white blood cells, and platelets). Plasma is primarily composed of water (over 90%), along with dissolved substances like proteins, electrolytes, nutrients, gases, and waste products. Plasma transports blood cells (red blood cells, white blood cells, and platelets) throughout the body. It carries nutrients, hormones, and proteins to various tissues and organs, and removes waste products. Plasma helps maintain body temperature and blood pressure. It also contains clotting factors, which are essential for blood clotting.

21. What is haemoglobin ?

- (a) Substance found in the leaves of plants
- (b) Substance found in the bone-marrow
- (c) Substance found in human blood
- (d) Secretion coming out from the pituitary gland

R.A.S./R.T.S. (Pre) 1992

Ans. (c)

Haemoglobin is the protein molecule in red blood cells that carries oxygen from the lungs to the body's tissues and returns carbon dioxide from the tissues back to the lungs.

22. Which of the following statements is NOT correct about haemoglobin?

- (a) It is red coloured
- (b) It is the carrier of oxygen from lungs to the cells
- (c) It is slightly acidic
- (d) It carries carbon dioxide away from the tissues to the lungs

U.P. B.E.O. (Pre) 2019

Ans. (c)

Haemoglobin is the red coloured, iron-containing complex protein in the red blood cells that carries oxygen from the lungs to the cells and returns carbon dioxide away from the tissues to the lungs. A healthy individual has 12-16 gms of haemoglobin in every 100 ml of blood. The pH value of haemoglobin solution is between 7.1 to 7.4 and hence, it is slightly alkaline, not acidic.

23. Read the following statements in connection with red blood corpuscles and with the help, the code given below find out the correct answer ?

1. They have iron
2. They provide red colour to the blood
3. They provide immunity against certain diseases
4. They are the carrier of oxygen in the blood

Code :

- (a) 1, 2 and 3
- (b) 2, 3 and 4
- (c) 1, 2 and 4
- (d) 1, 2 3 and 4

U.P.U.D.A./L.D.A. (Pre) 2001

U.P.P.C.S. (Pre) 2001

U.P.P.C.S. (Spl.) (Mains) 2004

U.P.P.C.S. (Pre) 2005

Ans. (c)

Red blood cells (RBCs) or red blood corpuscles also called erythrocytes are the most common type of blood cell and the vertebrate organism's principal means of delivering oxygen (O₂) to the body tissues—via blood flow through the circulatory system. RBCs take up oxygen in the lungs or gills and release it into tissues while squeezing through the

body's capillaries. The cytoplasm of erythrocytes is rich in haemoglobin, an iron-containing biomolecule that can bind oxygen and is responsible for the red colour of the cells.

24. Consider the following statements about haemoglobin :

1. It carries oxygen in the blood.
2. It is iron-containing compound.
3. It provides a immunity against certain diseases.
4. It imparts a red colour to the blood.

The correct statements out of these are :

- (a) 1, 2 and 3 (b) 1, 3 and 4
(c) 2, 3 and 4 (d) 1, 2 and 4

U.P. P.C.S. (Pre) 2016

Ans. (d)

See the explanation of above question.

25. The function of haemoglobin in body is

- (a) Transport of Oxygen (b) Destruction of Bacteria
(c) Prevention of Anaemia (d) Utilization of Iron

Uttarakhand P.C.S. (Pre) 2006

Ans. (a)

See the explanation of above question.

26. Which of the following natural substances contains iron (Fe)?

- (a) Chlorophyll (b) Collagen
(c) Keratin (d) Myoglobin

U.P. P.C.S. (Pre) 2020

Ans. (d)

Myoglobin (Mb) is an oxygen-binding heme protein found in the skeletal muscle tissue of vertebrates in general and in almost all mammals. It contains iron (Fe). It is distantly related to haemoglobin. Compared to haemoglobin, myoglobin has a higher affinity for oxygen and does not have cooperative binding with oxygen like haemoglobin does.

27. Myoglobin contains the metal –

- (a) Copper (b) Silver
(c) Gold (d) Iron

47th B.P.S.C. (Pre) 2005

Ans. (d)

See the explanation of above question.

28. In which of the following creatures haemoglobin is dissolved in the plasma–

- (a) Frog (b) Fish
(c) Human (d) Earthworm

I.A.S. (Pre) 1994

Ans. (d)

An earthworm is one of the animals that are classified in the Annelida phylum. In an earthworm, haemoglobin is dissolved in the plasma. The earthworm has a closed circulatory system in which blood is confined to blood vessels which re-circulates again and again to get maximum use of it.

29. Due to which the blood has red colour :

- (a) Plasma (b) Haemoglobin
(c) Haemocynin (d) W.B.C.

U.P.P.C.S. (Pre) 1990

Ans. (b)

The red colour of red blood cells in the blood is due to the combining of a protein known as haemoglobin with oxygen. Red blood cells are also called erythrocytes which are found only in the blood of vertebrates. There is no nucleus inside the red blood cells.

30. Red blood corpuscles (RBCs) has colour due to :

- (a) Cutin
(b) Chlorophyll
(c) Haemocynin
(d) Haemoglobin

Uttarakhand U.D.A./L.D.A. (Pre) 2003

Ans. (d)

See the explanation of above question.

31. What function does blood perform in the body?

- (a) Takes oxygen to all parts
(b) Maintains liquidity
(c) Helps in digestion
(d) Helps in erection

M.P.P.C.S. (Pre) 2010

Ans. (a)

Blood performs many important functions within the body including supply of oxygen to tissues (bound to haemoglobin, which is carried in red cells) and the supply of nutrients such as glucose, amino acids and fatty acids (dissolved in the blood or bound to plasma protein).

32. Average blood pressure of a human is –

- (a) 60/100 (b) 20/80
(c) 60/140 (d) 120/80

U.P.P.C.S. (Pre) 1993

Ans. (d)

The normal average blood pressure of human is 120/80 mm Hg. This means that systolic pressure around 120mm Hg and diastolic pressure should be around 80 mm Hg.

33. The blood pressure (systolic and diastolic) of a healthy man is –

- (a) 120 mm and 80 mm (b) 201 mm and 110 mm
(c) 90 mm and 60 mm (d) 85 mm and 55 mm

Uttarakhand P.C.S. (Pre) 2005

U.P.P.C.S. (Pre) 1991

Ans. (a)

See the explanation of above question.

34. Which of the following instrument measures blood pressure –

- (a) Spherometer (b) Anemometer
(c) Sphygmomanometer (d) Ammeter

48th to 52nd B.P.S.C. (Pre) 2008

Ans. (c)

A Sphygmomanometer is a device used to measure blood pressure, composed of an inflatable cuff to restrict blood flow and a mercury or mechanical manometer to measure the pressure. It is always used in conjunction with a means to determine at what pressure blood flow is just starting and at what pressure it is unimpeded. Manual sphygmomanometers are used in conjunction with a stethoscope.

35. Which of the following instruments is used for measuring blood pressure ?

- (a) Hydrometer (b) Multimeter
(c) Salinometer (d) Sphygmomanometer

Uttarakhand Lower Sub. (Pre) 2010

Ans. (d)

See the explanation of above question.

36. The blood pressure of an individual is 140mm Hg. In this statement, 'Hg' refers to –

- (a) Hydrogen (b) Mercury
(c) Hygrometer (d) Heliograph

U.P.P.C.S. (Pre) (Re. Exam) 2015

Ans. (b)

'Hg' is a symbol of Mercury, a chemical element. It is used in thermometers, barometers, sphygmomanometers and other devices. A sphygmomanometer is a device used to measure blood pressure. The person's blood pressure is measured in millimeters of mercury (mmHg). Normal blood pressure of human body is 120/80 mmHg.

37. When a person becomes older, his blood pressure generally –

- (a) Decreases (b) Increases
(c) Remains the same (d) Varies widely

Uttarakhand P.C.S. (Pre) 2007

Ans. (b)

The causes of high blood pressure are not known, but several factors and conditions may play a role in its development, including : Smoking, being overweight or obese, lack of physical activity, too much salt in the diet, too much alcohol consumption (more than 1 to 2 drinks per day), stress, older age, genetics, family history of high blood pressure, chronic kidney disease, adrenal and thyroid disorders etc.

38. Blood pressure of four person is given below :

Mrs. (X) – 90/60

Mr. (X) – 160/120

Mr. (Y) – 120/80

Mrs. (Y) – 140/100

Whose blood/pressure is normal in the following :

- (a) Mrs. (X) (b) Mr. (X)
(c) Mrs. (Y) (d) Mr. (Y)

I.A.S. (Pre) 1993

Ans. (d)

Here blood pressure of Mr (Y) is normal because normal blood pressure in humans is generally 120/80, i.e. systolic pressure 120 mmHg and diastolic pressure 80 mm Hg. Blood pressure of Mrs. (X) is less than normal while Mr. (X) and Mrs. (Y) have blood pressure more than the normal.

39. The blood pressure in our body is –

- (a) Lesser than the atmospheric pressure
(b) Greater than the atmospheric pressure
(c) Equal to the atmospheric pressure
(d) None of the above

Uttarakhand P.C.S. (Mains) 2006

Ans. (b)

The normal atmospheric pressure is 760 mm of Hg. But the normal human blood pressure is around 120/80 mmHg only. Actually when a doctor measures our blood pressure the measurement is done with respect to atmospheric pressure. This means that our blood pressure 120mm Hg more than that of atmospheric pressure of that place. On the other hand atmospheric pressure is measured with respect of vacuum, so the actual blood pressure of our body will be $760+120 = 880$ mm Hg with respect to vacuum.

40. Blood grouping was discovered by –

- (a) Louis Pasteur (b) William Harvey
(c) Robert Koch (d) Landsteiner

U.P.P.C.S. (Pre) 2003

U.P. Lower Sub. (Pre) 2004

Uttarakhand P.C.S. (Pre) 2007

Ans. (d)

Karl Landsteiner was an Austrian biologist and physician. He is noted for having first distinguished the main blood groups in 1900, having developed the modern system of classification of blood groups from his identification of the presence of agglutinins in the blood and having identified with Alexander S. Wiener, the Rhesus (Rh) Factor in 1937 (which was believed to be a similar antigen found in rhesus monkey red blood cells), thus enabling physicians to transfuse blood without endangering the patient's life.

41. Rh factor derives its name from a type of :

- (a) Ape (b) Human
(c) Monkey (d) Rat

U.P. U.D.A./L.D.A. (Mains) 2010

Ans. (c)

See the explanation of above question.

42. Blood group of an individual is determined by :

- (a) Presence of Haemoglobin
(b) Combination of WBC and RBC
(c) Shape of RBC
(d) None of the above

70th B.P.S.C. (Pre) (Re-Exam) 2024

Ans. (d)

Blood group is determined by the presence or absence of specific antigens (proteins) on the surface of red blood cells. They are like 'labels' that help the immune system identify different types of cells. These antigens are part of the ABO blood group system and the Rh system. The ABO system classifies blood into A, B, AB, and O types based on the presence or absence of A and B antigens.
Type A : Has A antigens on red blood cells.
Type B : Has B antigens on red blood cells.
Type AB : Has both A and B antigens on red blood cells.
Type O : Has neither A nor B antigens on red blood cells.
The Rh system, determined by the presence or absence of the RhD antigen, further divides these into positive (+) and negative (−) types.
Rh Positive (+) : Has the RhD antigen on red blood cells.
Rh Negative (−) : Does not have the RhD antigen on red blood cells.

43. The Red Blood Cells of a person with Blood group 'O' contains Agglutinin :

- (a) A and B both (b) Only B
(c) Neither A nor B (d) Only A

Rajasthan P.C.S. (Pre) 2023

Ans. (c)

Agglutinin is an antigen that causes the formation of agglutinins in the body and leads to agglutination, such as haemagglutination, which involves red blood cells (RBCs). The kind of agglutinogens present on the red blood cells helps determine the blood type of a person. If a person has blood type A, then the red blood cells exhibit agglutinogens A or antigens A. If the blood is of type B, the agglutinogens present are of type B. If the blood is of type AB, then both agglutinogens A and B are present. In blood type O, there are no agglutinogens on the surface of the red blood cells.

44. The blood group of universal donor is –

- (a) B (b) O
(c) A (d) AB

40th B.P.S.C. (Pre) 1995

43rd B.P.S.C. (Pre) 1999

M.P. P.C.S. (Pre) 2000

U.P.U.D.A./L.D.A. (Pre) 2003

Ans. (b)

Individuals with blood group 'O' are a universal donor because their red blood cells have neither A nor B antigens on their surface. So, the blood of a person having O group can be given to people with any blood group.

45. For human blood transfusion, which blood group is known as a universal donor?

- (a) B⁺ Group (b) O Group
(c) AB Group (d) A⁺ Group

U.P. P.C.S. (Pre) 2016

Ans. (b)

See the explanation of above question.

46. A person of which blood group can be a universal donor?

- (a) O (b) A
(c) B (d) AB

M.P.P.C.S. (Pre) 2015

Ans. (a)

See the explanation of above question.

47. Assertion (A) : People with AB blood groups are universal recipients.

Reason (R) : Red blood cell of blood group AB has no antigen and so agglutination does not occur with any other blood group.

Code :

- (a) both (A) and (R) are individually true, and (R) is the correct explanation of (A).

- (b) both (A) and (R) are individually true, but (R) is the correct explanation of (A).
 (c) (A) is true, but (R) is false.
 (d) (A) is true, but (R) is true.

M.P.C.S. (Pre) 2008

Ans. (c)

Individuals with blood group AB have both A and B antigens on the surface of their RBCs and their blood plasma does not contain any antibodies. Therefore an individual with type AB blood can receive blood from any group (with AB being preferable), but cannot donate blood to any group other than AB. They are known as universal recipients.

48. The universal acceptor blood group is :

- (a) A (b) B
 (c) AB (d) O
 (e) None of the above/More than one of the above

67th B.P.S.C. (Pre) 2022

Ans. (c)

See the explanation of above question.

49. Blood group AB can accept blood from a person of which blood group ?

- (a) A Only (b) B Only
 (c) AB Only (d) Any Group

Chhattisgarh P.C.S. (Pre) 2003

Ans. (d)

See the explanation of above question.

50. A person with 'AB' blood group is sometimes called a universal recipient because of the :

- (a) Lack of antigen in his blood
 (b) Lack of antibodies in his blood
 (c) Lack of both antigens and antibodies in his blood
 (d) Presence of antibodies in his blood

I.A.S. (Pre) 1995

Ans. (b)

See the explanation of above question.

51. A man having 'A' blood group met a serious road accident and the doctors advised for blood transfusion. The relatives were asked to donate blood, whose blood groups were found to be as follows :

- (i) Wife – 'O' (ii) Brother – 'AB'
 (iii) Son – 'A' (iv) Daughter – 'O'

Who among the above relations could donate blood to the injured man?

- (a) (i) and (ii) (b) (ii) and (iii)

(c) (ii), (iii) and (iv)

(d) (i), (iii) and (iv)

R.A.S./R.T.S.(Pre) 2013

Ans. (d)

Blood group A individuals have the A antigen on the surface of their RBCs and blood serum containing IgM antibodies against the B antigen. Therefore, group A individual can receive blood only from individuals of groups A or O (with A being preferable) and can donate blood to individuals with type A or AB.

52. A person meets with an accident and needs a blood transfusion but there is no time to check his blood group. Which of the following blood groups could be given to him?

- (a) O⁺ (b) O⁻
 (c) AB⁺ (d) AB⁻

U.P.C.S. (Pre) 2006

R.A.S./R.T.S. (Pre) 2013

Ans. (b)

There are four main blood groups defined by ABO system.

1. Blood group A – has A antigens on the RBC with anti-B antibodies in the plasma.
2. Blood group B – has B antigens with anti-A antibodies in the plasma.
3. Blood group AB – has both A and B antigens, but no antibodies.
4. Blood group O – has no antigens, but both anti-A and anti-B antibodies in the plasma.

Red blood cells sometimes have another antigen, a protein known as RhD antigen (Rh factor). If this is present, the blood group is RhD-positive. If it is absent, the blood group is RhD-negative. In most cases, RhD-negative blood (O⁻) can safely be given to anyone. It is often used in medical emergencies when the blood type isn't immediately known. It's safe for most users because it doesn't have any A, B or RhD antigens on the surface of the blood cells, and is compatible with every other A,B,O and RhD blood group.

53. A man whose blood group is not known meets with a serious accident and needs blood transfusion immediately. Which one of the blood groups mentioned below and readily available in the hospital will be safe for transfusion?

- (a) O, Rh⁻ (b) O, Rh⁺
 (c) AB, Rh⁻ (d) AB, Rh⁺

R.A.S./R.T.S. (Pre) 2013

I.A.S. (Pre) 2001

Ans. (a)

See the explanation of above question.

54. A married couple adopted a male child. A few years later, twin boys were born to them. The blood group of the couple is AB positive and O negative. The blood group of the three sons is A positive, B positive and O positive. The blood group of the adopted son is –

- (a) O positive
- (b) A positive
- (c) B positive
- (d) Cannot be determined on the basis of the given data

I.A.S. (Pre) 2011

Ans. (a)

As the blood group of the couple is AB⁺ and O⁻, twin boys were born to them will have A⁺ and B⁺ blood groups. Therefore, blood group of the adopted son is O⁺.

55. If the blood group of father is A and mother is 'O' then what will be the blood group of son –

- (a) B
- (b) AB
- (c) O
- (d) B, AB or O

I.A.S. (Pre) 1994

Ans. (c)

If among the parent blood group of the father is A and blood group of the mother is 'O' then the blood group of the son will be 'O' or 'A'. Bernstein in 1924 discovered that blood group A, B, O in humans represents genetic characteristic and develop according to Mendel's laws, in offsprings by getting genes from parents.

56. In transfusion, blood must be compatible not only in blood type but also in :

- (a) Rh factor
- (b) the number of white cells
- (c) the number of red cells
- (d) race of donor and recipient
- (e) None of the above/More than one of the above

67th B.P.S.C. (Pre) 2022

Ans. (a)

Blood is a special connective tissue consisting of a fluid matrix, plasma, and formed elements. The Rh antigen similar to one present in Rhesus monkeys (hence Rh), is also observed on the surface of RBCs in majority (nearly 80 percent) of humans. Such individuals are called Rh positive (Rh+ve) and those in whom this antigen is absent are called Rh negative (Rh-ve). An Rh-ve person, if exposed to Rh+ve blood, will form specific antibodies against the Rh antigens. Therefore, Rh factor should also be matched before transfusions.

57. Mother-foetus Rh blood type incapability problems can occur if the mother is and her foetus is :

- (a) Rh positive; Rh negative
- (b) Rh negative; Rh positive
- (c) Rh negative; Rh negative
- (d) Rh positive; Rh positive

R.A.S./ R.T.S. (Pre) 2021

Ans. (b)

Erythroblastosis fetalis, also known as hemolytic anaemia is a serious medical condition that most commonly results from maternal-fetal Rh blood type incompatibility. If a woman who is Rh-negative and a father who is Rh-positive conceive a child, the foetus may be Rh-positive. This can lead to Rh incompatibility and pregnancy complication and could cause the baby to develop severe anaemia and other problems.

58. What is the pH level of blood of a normal person?

- (a) 4.5 - 4.6
- (b) 6.45 - 6.55
- (c) 7.35 - 7.45
- (d) 8.25 - 8.35

U.P.P.C.S. (Mains) 2007

I.A.S. (Pre) 2008

M.P.P.C.S. (Pre) 2010

Ans. (c)

The pH value of the blood of a normal person is about 7.4. Blood pH is regulated to stay within the narrow range of 7.35 to 7.45, making it slightly basic or alkaline. Blood that has a pH below 7 is acidic, whereas blood pH above 7.45 is too basic.

59. The pH value of blood is –

- (a) 5.0
- (b) 6.4
- (c) 7.4
- (d) 8.0

Jharkhand P.C.S. (Pre) 2013

Ans. (c)

See the explanation of above question.

60. The pH value of the blood in human body is –

- (a) 6.4
- (b) 4.8
- (c) 7.4
- (d) 8.4
- (e) 6.2

Chhattisgarh P.C.S (Pre) 2013

Ans. (c)

See the explanation of above question.

61. An antigen is a substance which :

- (a) Destroys harmful bacteria
- (b) Is used to treat poisoning
- (c) Lowers body temperature
- (d) Stimulates formation of antibody

I.A.S. (Pre) 2001

Ans. (d)

An antigen is usually a foreign protein molecule (in some cases it is a polysaccharide) which enters the body fluids of an animal as part of an infectious agent and is capable of inducing the production of specific antibodies. An antibody is a molecule produced by animals in response to antigen and has the particular property of combining specifically with the antigen which induced its formation. It is a blood protein (globulin) which may make an infected animal immune to a foreign antigen. This response is called the immune response.

62. An antigen is a substance which –

- (a) Lowers body temperatures
- (b) Destroys harmful bacteria
- (c) Triggers the immune system
- (d) Is used as an antidote to poison

I.A.S. (Pre) 1997

Ans. (c)

See the explanation of above question.

63. What is the basic feature of antigen –

- (a) It induces formation of haemoglobin
- (b) It induces formation of antibodies
- (c) It destroys insulin
- (d) It acts against the antibodies

Jharkhand P.C.S. (Pre) 2003

Ans. (b)

See the explanation of above question.

64. Which of the following helps us in protecting from infection –

- (a) R.B.C.
- (b) W.B.C.
- (c) Blood Plasma
- (d) Haemoglobin

R.A.S./R.T.S. (Pre) 1999

Ans. (b)

White blood corpuscles (WBCs) are disease-fighting cells found in blood. When our blood is infected by any harmful bacteria or virus at any place in the body, white blood corpuscles reach there and eat up or destroy these harmful outsiders.

65. The main function of white blood cells is –

- (a) Transport of oxygen
- (b) Transport of carbon dioxide
- (c) To develop resistance towards disease
- (d) None of the above

Jharkhand P.C.S. (Pre) 2003

Ans. (c)

See the explanation of above question.

66. Antibodies are formed in blood plasma by which of the following –

- (a) Monocytes
- (b) Lymphocytes
- (c) Eosinophils
- (d) Neutrophils

Jharkhand P.C.S. (Pre) 2003

Ans. (b)

Lymphocytes produce antibodies in blood plasma and are responsible for inactivating poisonous substances. Lymphocytes are very small white blood cells with large nucleus, wander in the whole body and their number is large in total W.B.C. count (about 20-25%). Like an army, lymphocytes protect the body by co-ordinating with all parts of the immune system. Lymphocytes include natural killer cells, T cells and B cells which are the major cellular components of the immune response.

67. The immunity (defence) is mostly related with -

- (a) Lymphocytes
- (b) Monocytes
- (c) Red blood cells
- (d) Thrombocytes

R.A.S./R.T.S. (Pre) 1993

Ans. (a)

See the explanation of above question.

68. The most important cell type associated with the immunity of the body is :

- (a) RBCs
- (b) platelets
- (c) lymphocytes
- (d) More than one of the above
- (e) None of the above

68th B.P.S.C. (Pre) 2022

Ans. (c)

See the explanation of above question.

69. Which one of the following statements best describes the role of B cells and T cells in the human body?

- (a) They protect the body from environmental allergens.
- (b) They alleviate the body's pain and inflammation.
- (c) They act as immunosuppressants in the body.
- (d) They protect the body from the diseases caused by pathogens.

I.A.S. (Pre) 2022

Ans. (d)

The B and T lymphocytes (B and T Cells) protect the human body from the diseases caused by the pathogens. T cells and B cells are the most critical components of the adaptive immune system. These cells are essential for fighting diseases and play an important role in regulating hypersensitivity to harmless or 'self' antigens. T cells recognize viral antigens outside the infected cells, whereas B cells can recognize the surface antigens of bacteria and viruses directly.

They are involved in the acquired or antigen-specific immune response given that they are the only cells in the organism able to recognize and respond specifically to each antigenic epitope. The B Cells have the ability to transform into plasmocytes and are responsible for producing antibodies (Abs). Thus, T cells are involved in cell-mediated immunity, whereas B cells are primarily responsible for humoral immunity (related to antibody production in body fluids).

70. When antigens are introduced in the body through vaccination, antibodies are produced to fight off infection :

- (a) By both B and T cells (b) By only B cells
(c) By only T cells (d) None of the above

70th B.P.S.C. (Pre) (Re-Exam) 2024

Ans. (b)

B cells and T cells are a specific type of white blood cell called lymphocytes. Lymphocytes fight harmful invaders and abnormal cells, like cancer cells. T cells protect us by destroying pathogens and sending signals that help coordinate our immune system's response to threats. B cells make infection-fighting proteins called antibodies. When antigens are introduced through vaccination, B cells or B lymphocytes are the primary cells responsible for producing antibodies to fight off infection. B cells recognize and respond to antigens, triggering the production of antibodies. These antibodies then target and neutralize the specific antigen, helping to prevent or reduce the severity of the infection.

71. The viscous nature of human blood is due to –

- (a) Proteins in blood
(b) Platelets in plasma
(c) RBC and WBC in blood
(d) All of the above

Jharkhand P.C.S. (Pre) 2013

Ans. (a)

The primary determinants of blood viscosity are hematocrit and plasma viscosity. Plasma's viscosity is determined by water-content and macromolecular components, so these factors that affect blood viscosity are the plasma protein concentration and types of proteins in the plasma.

72. The diameter of white blood corpuscles in human body is, about :

- (a) 0.007 mm (b) 0.7 mm
(c) 0.07 mm (d) 0.0007 mm

Chhattisgarh P.C.S. (Pre) 2008

Ans. (a)

The number of leukocytes (WBCs) in the blood is often an indicator of disease. The normal white blood cell count is about 4,000–11,000 per micro litre of blood. They make up approximately 1% of the total blood volume in a healthy adult. An increase in the number of leukocytes over the upper limits is called leukocytosis, and a decrease below the lower limit is called leukopenia. The diameter of W.B.C. in human body is about 0.007 mm.

73. In pathology excess of white corpuscles in the blood is called

- (a) Anoxia (b) Leukemia
(c) Anaemia (d) Septicemia

Uttarakhand P.C.S. (Pre) 2007

Ans. (b)

Leukemia is a group of cancer that usually begins in the bone marrow and results in high numbers of abnormal white blood cells. These white blood cells are not fully developed and are called blasts or leukemia cells. Symptoms may include bleeding and bruising problems.

74. In which part of our body are formed Red Blood Cells?

- (a) Liver (b) Heart
(c) Kidney (d) Bone Marrow

Uttarakhand Lower Sub. (Pre) 2010

Uttarakhand U.D.A./L.D.A. (mains) 2006

Ans. (d)

Erythropoiesis is the development process by which new erythrocytes or Red Blood Cells are produced; it lasts about 7 days. Through this process, erythrocytes are continuously produced in the red bone marrow of large bones at a rate of about 2 million per second in a healthy adult (In the embryo, the liver is the main site of red blood cell production). The production can be stimulated by the hormone erythropoietin (EPO), synthesized by the kidney. Just before and after leaving the bone marrow, the developing cells are known as reticulocytes; these comprise about 1% of circulating red blood cells.

75. Formation of WBC and destruction of RBC takes place in –

- (a) Lymph gland (b) Spleen
(c) Pancrease (d) Liver

U.P.P.C.S. (Spl.) (Mains) 2008

Ans. (b)

The functional lifetime of an erythrocyte is about 100–120 days, during which time the erythrocytes are continually moved by the blood flow push (in arteries), pull (in veins) and a combination of the two as they squeeze through microvessels such as capillaries. The spleen is an organ found in virtually all vertebrates. Similar in structure to a large lymph node, it acts primarily as a blood filter. The spleen plays important role in regard to red blood cells (also referred to as erythrocytes) and the immune system. It removes old red blood cells and holds a reserve of blood, which can be valuable in case of haemorrhagic shock and also recycles iron. In the human adult, the bone marrow alone produces all of the RBCs, 60-70 percent of the WBCs (i.e. the granulocytes), and all of the platelets. The bone marrow and the lymphatic tissues, particularly the spleen, the thymus, and the lymph nodes, produce the lymphocytes (comprising 20-25 percent of WBCs).

76. Lymphocytes cells are formed in which part of human body?

- (a) Liver (b) Kidney
(c) Pancreas (d) Spleen

Jharkhand P.C.S. (Pre) 2010

Ans. (d)

See the explanation of above question.

77. In which organ of the human body are the lymphocytes formed?

- (a) Liver (b) Bone marrow
(c) Spleen (d) Pancreas
(e) None of the above/More than one of the above

65th B.P.S.C. (Pre) 2019

Ans. (e)

See the explanation of above question.

78. Arteries supplying blood to the heart are called :

- (a) Carotid arteries (b) Hepatic arteries
(c) Coronary arteries (d) Pulmonary arteries

I.A.S. (Pre) 1997

Ans. (c)

The vessels that deliver oxygen-rich blood to the myocardium are known as coronary arteries. The vessels that remove the deoxygenated blood from the heart muscle are known as cardiac veins.

79. With which of the physiological process thrombin is associated?

- (a) Excretion (b) Blood clotting

(c) Reproduction

(d) Growth

U.P.P.C.S. (Pre) 2011

Ans. (b)

Thrombin is an enzyme that presides over the conversion of a substance called fibrinogen to fibrin, which promotes blood clotting. Thrombin is a serine protease in humans which is encoded by the F_2 gene. Prothrombin (coagulation factor II) is proteolytically cleaved to form thrombin in the coagulation cascade, which ultimately results in the reduction of blood loss. Thrombin, in turn, acts as a serine protease that converts soluble fibrinogen into insoluble strands of fibrin as well as catalyzing many other coagulation-related reactions.

80. The enzyme which takes part in changing fibrinogen to fibrin when blood clots are formed–

- (a) Pepsin (b) Maltase
(c) Thrombin (d) Prothrombin

R.A.S./R.T.S. (Pre) 1999

Ans. (c)

See the explanation of above question.

81. Percentage of water in plasma is –

- (a) 60% (b) 70%
(c) 80% (d) 90%

Jharkhand P.C.S. (Pre) 2010

Ans. (d)

Plasma is the liquid portion of blood-protein-salt solution which is about 91 to 92% of water, and constitutes about 55% of blood volume. Plasma contains albumin (protein), fibrinogen and globulins (including antibodies).

82. With reference to the blood in a normal person, which one of the following statements is correct ?

- (a) Compared to arteries, veins are less numerous and hold less of the body's blood at any given time
(b) Blood cells constitute about 70 percent of the total volume of the blood
(c) White blood cells (WBC) are made by lymph nodes only
(d) The blood has more platelets than WBC

I.A.S. (Pre) 2002

Ans. (d)

Platelets help in the blood clotting process (or coagulation) by gathering at the site of an injury, sticking to the lining of the injured blood vessel and forming a platform on which blood coagulation can occur. Platelets are only about 20% of the diameter of red blood cells. The normal platelet count is about 150,000-350,000 per microlitre of blood, but since platelets are so small, they make up just a tiny